Vector Spaces

Recall from last class:

- **Theorem 3.3:** If A is an $n \times n$ matrix and E is an elementary matrix, then
$$\det(EA) = \det(E)\det(A),$$
and
$$\det(E) = \begin{cases} 1 & \text{if } E \text{ represents adding a multiple of a row to another row,} \\ -1 & \text{if } E \text{ represents interchanging two rows,} \\ r & \text{if } E \text{ represents scaling a row by a factor of } r. \end{cases}$$
- **Theorem 3.6:** If A and B are n by n matrices, then
$$\det(AB) = \det(A)\det(B).$$
- **Corollary:** A is invertible if and only if $\det(A) \neq 0$.
- **Theorem:** If A is a 2×2 matrix, then the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, then the volume of the parallelepiped is $|\det A|$.
- **Theorem:** Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be linear with associated matrix A. If S is a parallelogram in $\mathbb{R}^2$, then $\text{Area}(T(S)) = |\det A|\text{Area}(S)$.

Definition 1. *A real* **vector space** *is a non-empty set V on which two operations, scalar multiplication and addition, are defined subject to the following 10 axioms. We call elements in V vectors. Let $u, v, w \in V$ and $c, d \in \mathbb{R}$.*

1. $u + v \in V$.
2. $u + v = v + u$.
3. $(u + v) + w = u + (v + w)$.
4. *there exists a vector* 0_V *in* V *such that* $u + 0_V = u$.
5. *there exists a vector* $(-u)$ *in* V *such that* $u + (-u) = 0_V$.
6. $cu \in V$.
7. $c(u + v) = cu + cv$.
8. $(c + d)u = cu + du$.
9. $c(du) = (cd)u$.
10. $1 \cdot u = u$.

1. Determine if the following are vector spaces.

(a) $\mathbb{R}^n$ with the usual operations.

(b) $S = \{(y_0, y_1, \ldots) : y_k \in \mathbb{R} \text{ for all } k \geq 0\}$. Define addition component-wise and scalar multiplication by scaling every component.

(c) Let $V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 : x^2 + y^2 = 1 \right\}$ with the usual operations.

(d) Let $\mathbb{P}_n = \{a_0 + a_1 t + \cdots + a_n t^n : a_0, \ldots, a_n \in \mathbb{R}\}$ with operations defined by corresponding operations on polynomials.

(e) Let $M_{n \times n}$ be the set of $n \times n$ matrices with the operations coming from matrix addition and scaling a matrix.

(f) Let $W = \{(x, y, 1) : x, y \in \mathbb{R}\}$ with the usual operations.

(g) Let $Z = \{(a, b) : a, b \in \mathbb{R}\}$ and $(a, b) + (c, d) = (a + c, 0)$ and $r(a, b) = (ra, 0)$.

(h) Let $Y = \{(a, b) : a, b \in \mathbb{R}\}$ and $(a, b) + (c, d) = (a + c, b - d)$ and $r(a, b) = (ra, rb)$.

(i) Let $F = \{f : \mathbb{R} \to \mathbb{R} : f \text{ is a function}\}$ with operations coming from the usual operations on functions.

Definition 2. *A* **subsapce** *of a vector space V is a subset of V such that*

(a) $0_V \in H$.

(b) *H is closed under vector addition; i.e. if $u, v \in H$, then $u + v \in H$.*

(c) *H is closed under scalar multiplication; i.e. if $c \in \mathbb{R}$ and $u \in H$, then $cu \in H$.*

2. Let V be a vector space.

(a) Is $H = \{0_V\}$ a subspace of V?

(b) Suppose $V = \mathbb{R}^3$, then is $\mathbb{R}^2$ a subspace of V?

Theorem 4.1 If $v_1, \ldots, v_p \in V$, then $\text{Span}\{v_1, \ldots, v_p\}$ is a subspace of V. We call it the subspace generated by or spanned by $v_1, \ldots, v_p$.

3. Prove the Theorem.

4. Consider the vector space of polynomials of degree 2 or less. We denote this vector space by $\mathbb{P}_2$.

(a) Is the set of polynomials of degree 2 a subspace?

(b) Let $u = 1 + t$, $v = t$, and $w = 1 + t + 2t^2$. Is $w \in \text{Span}\{u, v\}$?

5. Let W_1 and W_2 be subspaces of a vector space V.

(a) When is $W_1 \cap W_2$ a subspace of V?

(b) When is $W_1 \cup W_2$ a subspace of V?

6. Jonathan's question: If v is a vector space, then is $(-1)u = -u$?

Vector Spaces – Answers / Solutions

1. (a) We have previous seen that $\mathbb{R}^n$ with vector addition and scalar multiplication satisfies the 10 axioms, and hence it is a vector space.

(b) S is a vector space.

(c) V is not a vector space, since adding two elements of V doesn't always yield another element in V.

(d) $\mathbb{P}_n$ is a vector space.

(e) $M_{n\times n}$ is a vector space.

(f) W is not closed under scalar multiplication or vector addition, so it is not a vector space.

(g) Z does not have a zero vector, so it is not a vector space.

(h) The addition operation in Y is not commutative, so Y is not a vector space.

(i) F is a vector space.

2. (a) Yes, H is a subspace. We can prove it be checking the 3 conditions to be a subspace.
(a) Since $0_V \in H$, we know (a) holds.
(b) By vector space axiom 4, we know $0_V + 0_V = 0_V$. Thus H is closed under vector addition.
(c) Let $c \in \mathbb{R}$, then we want to prove $c0_V \in H$. By axiom 7, $c0_V = c(0_V + 0_v) = c0_V + c0_V$. Now subtracting $c0_V$ from both sides (axiom 5) yields $c0_V = 0_V$, and hence H is closed under scalar multiplication.

(b) No, $\mathbb{R}^2$ is not a subset of V, so it cannot be a subspace of V.

3. *Proof.* We need to check the three subspace conditions.

(a) Since $0_V = 0 \cdot v_1 + \cdots + 0 \cdot v_p$, we know $0_V \in \text{Span}\{v_1, \ldots, v_p\}$.

(b) Let $u, v \in \text{Span}\{v_1, \ldots, v_p\}$. By definition of the span, we know there exists scalars $c_1, \ldots c_p, d_1, \ldots, d_p \in \mathbb{R}$ such that

$$u = c_1v_1 + \cdots + c_pv_p$$

and

$$v = d_1v_1 + \cdots + d_pv_p.$$

By vector space axioms (2) and (8), we get

$$\begin{aligned} u + v &= c_1v_1 + \cdots + c_pv_p + d_1v_1 + \cdots + d_pv_p \\ &= c_1v_1 + d_1v_1 + \cdots + c_pv_p + d_pv_p \\ &= (c_1 + d_1)v_1 + \cdots + (c_p + d_p)v_p. \end{aligned}$$

Thus $u + v$ is a linear combination of $v_1, \ldots, v_p$ and $u + v \in \text{Span}\{v_1, \ldots, v_p\}$.

(c) Let $c \in \mathbb{R}$ and $u \in \text{Span}\{v_1, \ldots, v_p\}$. Again by definition of span, there exists $c_1, \ldots, c_p \in \mathbb{R}$ such that $u = c_1v_1 + \cdots + c_pv_p$, and using axioms 7 and 9, we get

$$cu = c(c_1v_1 + \cdots + c_pv_p) = (cc_1)v_1 + \cdots + (cc_p)v_p.$$

Therefore $cu \in \text{Span}\{v_1, \ldots, v_p\}$. $\square$

4. (a) No. The zero vector in $\mathbb{P}_2$ is the zero polynomial; namely, $0_{\mathbb{P}_n} = 0+0\cdot t+0\cdot t^2$. However, the zero polynomial is not of degree 2, so the set of polynomials of degree 2 fails the first condition to be a subspace.

(b) No. For w to be in the span of u and v, we would need to find scalars c_1, c_2 such that $w = c_1u + c_2v$, or equivalently $1 + t + 2t^2 = c_1(1 + t) + c_2t$. This is impossible since the left hand side is quadratic and the right hand side is linear. Thus $w \notin \text{Span}\{u, v\}$.

5. (a) $W_1 \cap W_2$ is a subspace of V.

Proof. **(a)** Since W_1 and W_2 are both subspaces of V, we know $0_V \in W_1$ and $0_V \in W_2$. Thus $0_V \in W_1 \cap W_2$.

(b) Let $u, v \in W_1 \cap W_2$. Since $u, v \in W_1$ and W_1 is a subspace, we know $u + v \in W_1$. Since $u, v \in W_2$ and W_2 is a subspace, we know $u + v \in W_2$. Thus $u + v \in W_1 \cap W_2$.

(c) Let $c \in \mathbb{R}$ and $u \in W_1 \cap W_2$. Since W_1 is a subspace, we get $cu \in W_1$. Since W_2 is a subspace, we get $cu \in W_2$. Thus $cu \in W_1 \cap W_2$.

Thus $W_1 \cap W_2$ is a subspaace of V. $\square$

(b) First note that generally $W_1 \cup W_2$ is not subspace of V. For instance take $V = \mathbb{R}^2$ and $W_1 = \text{Span}\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}$ and $W_2 = \text{Span}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}$. Then $\begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix} \notin W_1 \cup W_2$.

Claim: $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Proof. We prove each implication directly.

($\Longrightarrow$) Let $x \in W_1$ and $y \in W_2$. Since $W_1 \cup W_2$ is a subspace, we know $x + y \in W_1 \cup W_2$.

Case 1: $x + y \in W_1$.

Now $y = (x + y) - x$, so y is the sum of two elements in W_1. Since W_1 is a subspace, we get $y \in W_1$. Thus $W_2 \subseteq W_1$.

Case 2: $x + y \in W_2$.

Here $x = (x + y) - y$, so x is the sum of two elements in W_2. Since W_2 is a subspace, we get $x \in W_2$. Thus $W_1 \subseteq W_2$.

Therefore, if $W_1 \cup W_2$ is a subspace of V, then $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

($\Longleftarrow$) If $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$, and hence $W_1 \cup W_2$ is a subspace of V. If $w_2 \subseteq W_1$, then $W_1 \cup W_2 = W_1$, and hence $W_2 \cup W_2$ is a subspace of V. In either case, $W_1 \cup W_2$ is a subspace of V. $\square$

1. **Jonathan's Claim:** If V is a vector space, then $(-1)u = -u$.

Proof. Note that by definition $-u$ is the additive inverse of u.

Subclaim: additive inverses are unique.

Proof of subclaim: Suppose x and y are both additive inverses for u; namely $u + x = 0$ and $u + y = 0$. Now we have

$$x = 0 + x = (u + y) + x = y + (u + x) = y + 0 = y$$

where we used axioms 2, 4, 3.

Note

$$u + (-1)u = (1 + (-1))u = 0 \cdot u = 0.$$

Thus $(-1)u$ is the additive inverse of u; namely, $(-1)u = -u$.

$\square$